everywhere around it, fading in magnitude as you get farther away from the charge. If we shake an electron, oscillating it up and down, the field oscillates along with it, in ripples that gradually spread out from its location. This is electromagnetic radiation, or "light" for short. Every time we heat up a material to sufficient temperature, electrons in its atoms start to shake, and the material begins to glow. This is known as *black-body radiation*, and every object with a uniform temperature gives off a form of blackbody radiation.

Red light corresponds to slowly oscillating, low-frequency waves, while blue light is rapidly oscillating, high-frequency waves. Given what physicists knew about atoms and electrons at the turn of the century, they could calculate how much radiation a blackbody should emit at every different frequency, the so-called blackbody spectrum. Their calculations worked well for low frequencies, but became less and less accurate as they went to higher frequencies, ultimately predicting an infinite amount of radiation coming from every material body. This was later dubbed the "ultraviolet catastrophe," referring to the invisible frequencies even higher than blue or violet light.

Finally in 1900, German physicist Max Planck was able to derive a formula that fit the data exactly. The important trick was to propose a radical idea: that every time light was emitted, it came in the form of a particular amount—a "quantum"—of energy, which was related to the frequency of the light. The faster the electromagnetic field oscillates, the more energy each emission will have.

In the process, Planck was forced to posit the existence of a new fundamental parameter of nature, now known as *Planck's constant* and denoted by the letter *h*. The amount of energy contained in a quantum of light is proportional to its frequency, and Planck's constant is the constant of proportionality: the energy is the frequency times *h*. Very often it's more convenient to use a modified version $\hbar$, pronounced "h-bar," which is just Planck's original constant *h* divided by 2π. The appearance of Planck's constant in an expression is a signal that quantum mechanics is at work.

Planck's discovery of his constant suggested a new way of thinking about physical units, such as energy, mass, length, or time. Energy is measured in units such as ergs or joules or kilowatt-hours, while frequency is measured in units of 1/time, since frequency tells us how many times something happens in a given amount of time. To make energy proportional